AAFNet: An Asymmetric Feature Encoding and Active Energy Modulation Network for Multimodal Remote Sensing Forest Fire Detection

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***Abstract*—Forest fires pose severe ecological threats, yet dedicated multimodal architectures for such complex environments remain largely underexplored. Generic multimodal methods struggle to decouple heterogeneous physical properties, resist feature contamination during dynamic fusion, and preserve the fragile boundaries of microfires against deep semantic smoothing. To address these limitations, we propose an asymmetric feature encoding and active energy modulation network (AAFNet) for multimodal remote sensing forest fire detection. First, an asymmetric feature encoding is designed at the source. Within this architecture, the geometry-aware anchor (GAA) module is proposed to adaptively fit the nonrigid boundaries of smoke via decoupled deformable sampling. In parallel, the thermal-aware salient (TAS) module performs hierarchical wide-range perception operators to capture long-range thermal gradients and suppress radiation artifacts. Second, an energy-guided modulation (EGM) mechanism is formulated to dynamically allocate modality weights by quantifying instantaneous energy responses. This active regulation suppresses background interference and improves feature purity during fusion. Third, a decoupled difference reconstruction (DDR) unit is developed to explicitly reconstruct residual differences between adjacent feature hierarchies. This compensation maximizes the preservation of fine-grained textures for early stage fires. Extensive experiments demonstrate the superiority of our approach. The proposed model achieves an mAP@50 of 75.2%, an mAP@50–95 of 52.1% on forest fire datasets with only 2.531 M parameters, which comprehensively outperforms state-of-the-art methods in both detection accuracy and deployment parameters. Moreover, cross-dataset evaluation further demonstrates the generalization capability of AAFNet under diverse environmental distributions.**

***Index Terms*—Energy-guided modulation (EGM), forest fire detection, multimodal object detection, remote sensing imagery.**

I. Introduction

FOREST fires pose a severe threat to global ecosystems and the carbon cycle. Driven by extreme climate conditions, their frequency and scale are increasing sharply [1], [2], [3]. Because early stage fires are often concealed within complex terrain and spread in a highly nonlinear manner, missing the "golden window" for intervention can easily trigger uncontrollable disasters.

Traditional forest fire monitoring relies on physical sensors [4], satellite remote sensing [5], [6], [7], or fixed surveillance [8]. However, ground-based sensors may suffer from deployment blind spots, while the field of view of fixed cameras can be obstructed by complex terrain. Satellite imagery also provides limited characterization of local fire details, such as flame regions, smoke morphology, and boundary structures. To overcome these limitations, UAVs enable flexible, close-range aerial inspection of suspected fire areas from adjustable viewing distances and angles. Consequently, accurate detection of flames and smoke from onboard imagery is crucial for autonomous forest fire warning.

Traditional visual methods for forest fire detection rely primarily on manual feature design. Early studies often constructed models using explicit rules based on low-level visual characteristics such as color and texture [9]. Other approaches applied machine learning algorithms such as support vector machines to improve classification reliability [10]. However, the reliance on handcrafted features not only demands substantial domain expertise but also yields rigid representations. As a result, their capacity to reliably decouple true fire targets from unconstrained wildland scenarios is restricted.

Subsequently, the rapid evolution of deep learning has propelled methodological innovations in forest fire detection, offering robust, automatic feature extraction while overcoming the limitations of traditional handcrafted methods. Kong et al. [11] proposed an attention-based dual-encoding network, which integrated spatial and semantic units to

Received 11 May 2026; revised 22 September 2026; accepted 2 October 2026. Date of publication 6 October 2026; date of current version 15 October 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62266049 and Grant 62576303, in part by the Research and Achievement Cultivation and Teaching Reformation Project on Graduate Education at Yunnan University under Grant JG-202407, in part by the "Intelligent Visual Perception" Supervisor Team for Graduate Students in Yunnan Province, and in part by the 2024 China Association for Science and Technology (CAST) Young Talents Support Program for Doctoral Students. *(Corresponding author: Haiyan Li.)*

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Digital Object Identifier 10.1109/TGRS.2026.3740840

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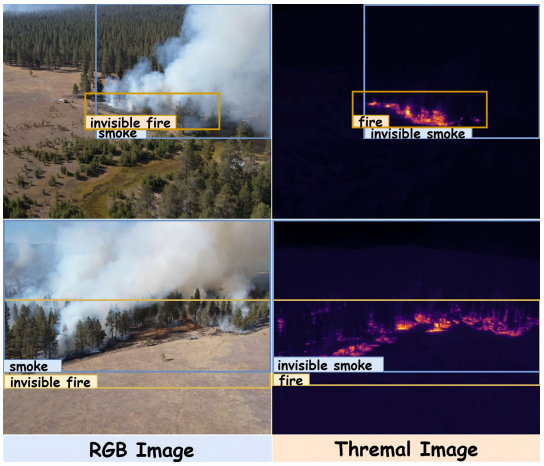


Fig. 1. Visual comparison of single-modality limitations in complex wildland fire scenarios. The RGB images (left column) clearly depict smoke, while early-stage fire sources may be obscured. In contrast, the corresponding TIR images (right column) reveal active fire regions despite smoke occlusion but provide limited information about smoke morphology.

improve the extraction of multiscale context and edge features for early small flames. Chen et al. [12] proposed the EMFFS-DET model based on multitask learning to improve local feature extraction and the suppression of complex background noise. However, both models rely on conventional local perception paradigms that fail to capture the physical properties of fire and smoke, and they are not adapted to irregular boundaries while misdetecting macroscopic thermal diffusion gradients. Wang et al. [13] proposed the FireSmoke YOLO model, which introduces fast funnel spatial pyramid pooling and dynamic snake convolution to improve adaptability to irregular fire shapes and enhance local feature extraction. Khan et al. [14] developed a cross-module attention network that utilizes a squeezing and expansive path alongside multiple-scale feature selection to improve the localization of fire features. Although existing methods improve local and multiscale fire representations, cross-level discrepancies remain underexplored. These discrepancies often retain structural cues weakened by semantic abstraction, making explicit modeling important for preserving weak early fire signals.

Despite these architectural advances, relying on single-modality paradigms leaves current methods [15], [16], [17] fundamentally vulnerable to dynamic wildland occlusions. As shown in Fig. 1, visible images preserve rich texture and smoke morphology but are vulnerable to smoke occlusion and illumination variations. Thermal infrared (TIR) images can highlight obscured high-temperature fire sources, yet provide limited smoke structure and are susceptible to interference from hot backgrounds. These differences indicate that the two modalities have distinct representation characteristics, and their effectiveness varies with environmental conditions. Therefore, multimodal fire detection should not only exploit complementary information but also adopt modality-specific representations and adaptively regulate modality contributions according to the current scene.

Such variations in modality effectiveness are related to robustness under distribution shifts. Sun et al. [18], [19] explored augmentation-based and weakly decoupled domain generalization, alongside pseudo-label-guided dual-classifier domain adaptation [20]. These fault diagnosis studies offer valuable insights into preserving discriminative information under changing conditions, with implications for multimodal forest fire perception under occlusion and environmental interference.

To summarize, the development of robust multimodal forest fire detection is currently hindered by three critical technical obstacles.

1) *Heterogeneity of Cross-Modal Physical Responses:* Smoke and fire sources in forest fire scenes exhibit markedly different spatial responses across modalities. In visible images, smoke often appears as a nonrigid structure with blurred boundaries, continuous deformation, and large-scale diffusion. In contrast, fire sources in TIR images are mainly characterized by continuous thermal radiation distributions and spatial temperature gradients. These differences in morphology, response range, and spatial distribution can cause mismatched cross-modal representations and weaken modality-specific discriminative information.
2) *Scene-Driven Fluctuations in Modality Effectiveness:* Complex wildland environments can continuously alter the effectiveness of information provided by RGB and TIR modalities. Dense smoke, occlusion, and illumination changes may substantially weaken target structures in visible images, while sun-heated vegetation, rocks, and other warm backgrounds may produce strong interference responses in TIR images. Therefore, neither modality maintains a fixed advantage across different scenes. When modality effectiveness changes, abnormal responses from the degraded modality can further interfere with cross-modal representations.
3) *Fine-Grained Information Weakening Caused by Deep Semantic Enhancement:* Small fire spots, irregular flame boundaries, and weak smoke edges rely heavily on fine-grained spatial information. As feature hierarchies deepen, stronger semantic representations can gradually weaken local structural responses. The resulting cross-level response discrepancies contain information about how target details change during hierarchical propagation. However, conventional multiscale aggregation often does not explicitly exploit these discrepancies, which further dilutes fine-grained cues.

To overcome these obstacles and advance the state of the art, we propose an asymmetric feature encoding and active energy modulation network (AAFNet) for multimodal remote sensing forest fire detection. Our contributions are threefold.

1) We propose an asymmetric feature encoding network to intrinsically decouple the heterogeneous feature responses across modalities at the extraction source. Within this architecture, the geometry-aware anchor (GAA) module designs decoupled deformable sampling to adaptively fit the nonrigid boundaries of smoke, and constructs an orthogonal spatial-anchoring mechanism to restore the global coordinate sensitivity

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of microtargets. In parallel, the thermal-aware salient (TAS) module captures long-range thermal gradients and suppresses radiation artifacts via hierarchical wide-range perception operators, achieving deep decoupling of modality-specific features.

2) An energy-guided modulation (EGM) mechanism is constructed to alleviate cross-modal interference caused by scene-dependent modality degradation in complex wildland environments. By quantifying the instantaneous energy response intensity of feature tensors, modality weights are dynamically allocated and effective perceptual components are adaptively compensated. When a single modality degrades, the active regulation suppresses its interfering responses, thereby reducing feature contamination during cross-modal integration.
3) Furthermore, a decoupled difference reconstruction (DDR) unit is designed for multiscale aggregation to alleviate the weakening of fine-grained structural information in fire targets during deep semantic representation. DDR explicitly models the residual discrepancies between adjacent feature hierarchies and reconstructs the structural cues embedded in these differences. It further adaptively coordinates the contributions of semantic, detail, and difference-enhanced information, thereby effectively preserving fine-grained fire representations during multiscale aggregation.

II. Related Work

A. Approaches to Forest Fire Detection

Recent years have witnessed significant advancements in forest fire detection, with state-of-the-art algorithms striving to achieve robust feature extraction. For instance, Wang et al. [21] utilized a dilation repconv cross stage partial network alongside a global mixed attention mechanism to extract multiscale features in parallel, thereby compensating for information loss caused by extreme illumination. Li et al. [16] introduced a cross-stream feature interaction mechanism to partition feature maps along the channel dimension for precise alignment and aggregation, thereby significantly bolstering cross-tier feature interactivity. Sharma and Kanwal [22] constructed a multistage classification detector built upon a lightweight MobileNet-V2 backbone, improving recognition efficiency in dynamic scenes by fusing shallow textures with deep semantics. Furthermore, Khan et al. [23] proposed a multiattention fire network that cascades channel and spatial attention modules within a ConvNeXt-T architecture to significantly bolster the model's ability to suppress background distractors. Wang et al. [24] further combined contextual partial cross features with dynamic multiscale fusion to enhance the representation of small fire and smoke targets. Despite these improvements, existing methods remain constrained by the paradigm of homogeneous feature extraction. Specifically, they rely on identical network branches and regular convolutional grids to indiscriminately process fire and smoke, ignoring their profound physical heterogeneity—such as fire's radiating thermal gradients versus smoke's nonrigid fluidity. Lacking tailored perception mechanisms, these rigid grids inevitably absorb environmental noise, leading to severe feature entanglement.

To overcome the limitations of homogeneous feature extraction and address the drastic scale variations of early fires, recent studies have developed differentiated perception pathways and refined multiscale aggregation architectures. For instance, Wang et al. [25] proposed a dual-pathway network that separates coarse- and fine-grained perception routes. Combined with a multidimensional interactive feature pyramid network, this approach effectively enhances decoupling and fusion. Targeting the highly variable morphology of smoke, Li et al. [26] designed a ghost module-based feature pyramid network to bolster the cross-layer representation of smoke features across different scales. Zhao et al. [27] introduced an unsupervised reconstruction mechanism based on VQ-VAE to explicitly measure structural similarity differences during feature reconstruction, effectively filtering out complex environmental noise for precise fire discrimination. Galván-Hernández et al. [28] constructed a hybrid spatial–temporal framework that combines maximum intensity projection with pixel-wise variance analysis to explicitly isolate incipient combustion dynamics from background thermal distractors. Additionally, Feng and Sun [29] incorporated dense connections and multiscale spatial feature pooling to facilitate the flow of local details across network hierarchies. However, these methods inevitably exhibit profound structural limitations when managing cross-layer feature intersections. Specifically, they tend to overemphasize the connection topology of macroscopic fusion pathways while neglecting the intrinsic representational capacity of feature transformation units. Consequently, during the transmission and intersection of deep and shallow features, these architectures fail to resist the semantic smoothing effect induced by deep networks.

In summary, to fundamentally overcome the dual bottlenecks of homogeneous feature extraction and multiscale feature intersection, we design an asymmetric perception architecture tailored to the physical heterogeneity of fire and smoke. This enables the precise decoupling of nonrigid geometric morphologies and thermal radiation gradients at the source. Furthermore, a DDR unit is proposed to dynamically compensate for interlayer residual features while effectively restoring the high-frequency textures and edge details of fire targets.

B. Multimodal Object Detection Methods

Recent years have witnessed significant advancements in multimodal object detection. Jiang et al. [30] introduced the union-modal attention and cross-modal attention modules to model the long-range dependencies between visible and TIR sequences, thereby adaptively aggregating multimodal global features. Shen et al. [31] put forward an iterative cross-modal feature enhancement mechanism to dynamically refine dual-modality complementary features from a global perspective via parameter-shared cross-attention computations. Bao et al. [32] proposed graph interaction and gated attention fusion mechanisms to deeply mine long-range dependencies among multimodal features along the channel and spatial dimensions. However, these methods dedicated to modeling

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global long-range dependencies inevitably introduce dense cross-attention matrix computations and cumbersome graph topology interactions. Zeng et al. [33] advanced the fusion focus module to dynamically discriminate spatial response patterns across modalities, thereby achieving the precise fusion of visible and infrared information. Chen et al. [34] devised dual semantic and spatial enhancing modules to dynamically redistribute cross-modality feature weights along the channel and pixel dimensions, thereby suppressing mutual interference during dual-modality fusion. Li et al. [35] proposed a multimodal and multiscale cross-compensation mechanism to explicitly model feature differences across modalities and dynamically compensate adjacent-level features. However, these feature-level compensation and redistribution mechanisms inherently over-rely on mathematical tensor mappings, severely neglecting the essential differences in the underlying imaging mechanisms of heterogeneous sensors. While effective for standard tasks, current fusion paradigms face critical bottlenecks in forest fire detection since nascent fires grow irregularly and often blend into complex thermal environments.

To bridge this gap, recent studies have sought to develop dedicated multimodal fire detection strategies that explicitly accommodate the distinctive multisource evolutionary patterns of forest fires. Xu et al. [5] integrated infrared and smoke signals at the subpixel scale to reduce omissions under weak radiation. Zhang et al. [36] further developed an STS-RNN to jointly model spatial, temporal, and spectral information from Himawari-8 observations, improving near-real-time wildfire discrimination. Subsequent work [37] integrated selected spectral, spatial, and temporal features from Himawari-8/9 and constructed separate models for early and continuous wildfire detection. More recently, a dynamic context-adaptive fusion method [38] combined local background statistics, spectral–temporal anomaly information, and spatiotemporal consistency constraints to achieve 10-min-level near-real-time wildfire monitoring. These studies effectively exploit the wide coverage and high revisit frequency of geostationary satellites for rapid regional wildfire monitoring. However, their main focus is fire-point discovery and spatiotemporal–spectral anomaly analysis, whereas close-range UAV observations require finer object-level representations of flames and smoke. Beyond satellite remote sensing, multimodal information has been applied to close-range forest fire perception. Jin et al. [39] proposed a multimodal smoke risk estimation network to integrate visual smoke features with environmental risk factors for lightweight forest fire early warning. Niu et al. [40] combined UAV visible and infrared observations for small-scale fire detection. These studies extend multimodal forest fire perception beyond satellite monitoring and demonstrate the value of complementary information from heterogeneous sources. The value of complementary representations is further illustrated by Sun et al. [41], who combined time–frequency fusion with latent representation prompts for open-set fault recognition. However, they mainly focus on the association or fusion of different information sources, while scene-dependent changes in the contributions of visual modalities remain insufficiently modeled.

To address these limitations, we propose the EGM mechanism. By explicitly capturing modality-specific energy distributions, this mechanism provides active guidance for feature interaction to achieve precise refinement of heterogeneous information with minimal computational cost.

III. Methodology

The overall architecture of the proposed multimodal forest fire detection network is illustrated in Fig. 2. The framework adopts an asymmetric dual-stream design to process visible and infrared images simultaneously. To address the physical differences between modalities, an asymmetric feature encoding network is constructed to intrinsically decouple heterogeneous properties at the extraction source. Specifically, the visible stream integrates the GAA to adaptively fit the nonrigid boundaries of fire and smoke. In parallel, the infrared stream utilizes the proposed TAS module to capture long-range thermal gradients and suppress radiation artifacts.

To integrate these decoupled representations, an EGM bridges the two processing streams across multiple network stages. This mechanism quantifies instantaneous energy responses to dynamically allocate modality weights. It actively suppresses background interference caused by dynamic signal-to-noise ratio inversions and improves feature purity during fusion. Following the cross-modal integration, the multiscale features are fed into a specialized aggregation neck. Here, our DDR explicitly calculates and reconstructs residual differences between adjacent feature hierarchies. This compensation prevents the smoothing of microtargets by deep semantic abstraction. Furthermore, it maximizes the preservation of fine-grained textures for early stage fires and outputs robust multiscale features to the final detection heads.

A. GAA Module

In early stage forest fire scenarios, visible images provide rich morphological cues for smoke perception. However, smoke regions usually exhibit nonrigid, anisotropic, and weakly bounded structures, making their contours difficult to describe using regular-grid aggregation. Moreover, low-altitude clouds, haze, and forest canopies often present cloudmist-like appearances that are visually similar to smoke, causing purely local responses to be easily contaminated by background structures. To address these issues, we propose the GAA module for the visible-light stream, as illustrated in Fig. 3. The primary objective of GAA is to enhance smoke-boundary representation by coupling local geometry-adaptive perception with macro-scene anchoring.

For local smoke-boundary perception, GAA constructs a learnable structural sampling field to adapt feature aggregation to the irregular morphology of smoke regions. Given an input feature map X, a near-square reference lattice P_n is defined around each spatial position p_0, and a learnable perturbation field Δp is predicted to generate geometry-aware sampling coordinates

$$p = p_0 + P_n + \Delta p. \tag{1}$$

Different from fixed convolutional grids, this formulation allows the sampling trajectory to be adjusted according to

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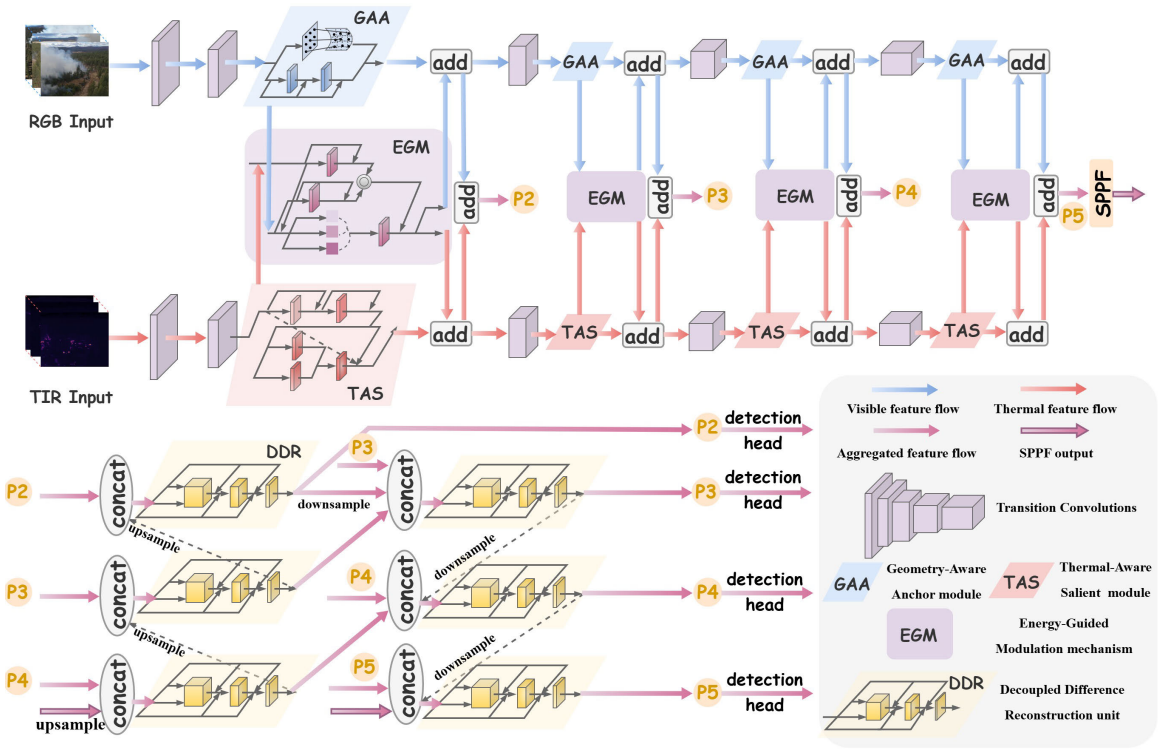


Fig. 2. Overall architecture of the proposed AAFNet. Representative GAA, TAS, EGM, and DDR modules are expanded to show their internal structures, while repeated instances are represented by compact blocks with consistent colors.

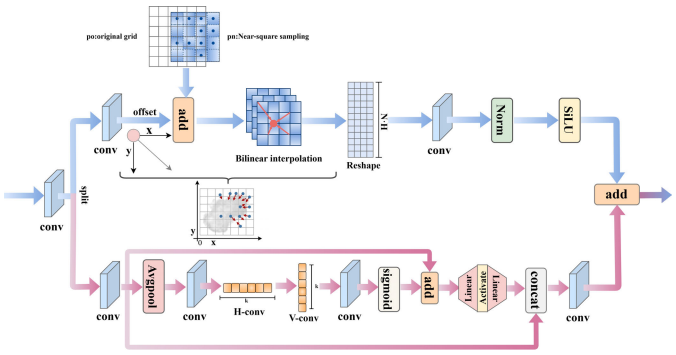


Fig. 3. Overall architecture of the GAA module.

local smoke contour variations, thereby reducing the mismatch between rigid receptive fields and nonrigid smoke structures. The sampled responses are then reorganized along the spatial dimension to form a continuous structural sequence. A lightweight directional projection is further adopted to aggregate the reordered responses and produce the geometry-enhanced feature

$$Y_{geo} = \sigma\left(\mathcal{B}\left(\mathcal{W}_{N\times 1} * \mathrm{Reorg}\left(X_s\right)\right)\right) \tag{2}$$

where X_s denotes the sampled feature responses, Reorg$(\cdot)$ represents the spatial reorganization operation, $\mathcal{W}_{N\times 1}$ is the directional projection kernel, $\mathcal{B}(\cdot)$ denotes batch normalization, and $\sigma(\cdot)$ is the SiLU activation function. Through this process, the visible stream is encouraged to preserve contour continuity for irregular smoke plumes, rather than relying on fixed local neighborhoods.

Although geometry-adaptive perception improves the fitting of smoke boundaries, local structural responses alone may still be confused by cloudmist-like backgrounds and large homogeneous forest regions. Therefore, GAA introduces a macro-scene anchor branch to provide contextual constraints for smoke discrimination. Specifically, the input feature X is first smoothed to suppress unstable high-frequency background fluctuations. Then, two orthogonal context encoding operations are successively applied along the horizontal and vertical directions to capture wide-area structural dependencies. The resulting macro-anchor map is formulated as

$$M = \sigma\left(\mathcal{W}_2 * \left(\mathcal{W}_{k\times 1} * \left(\mathcal{W}_{1\times k} * \left(\mathcal{W}_1 * \mathcal{P}_{avg}\left(X\right)\right)\right)\right)\right) \tag{3}$$

where $\mathcal{P}_{avg}(\cdot)$ denotes average pooling, $\mathcal{W}_{1\times k}$ and $\mathcal{W}_{k\times 1}$ encode orthogonal contextual dependencies, and $\mathcal{W}_1$ and $\mathcal{W}_2$ are point-wise projections. This decomposition provides a wide-area structural prior with linear complexity with respect to the kernel size, helping the network suppress cloud mist and canopy interference during smoke-boundary extraction.

The generated macro-anchor map is used to calibrate the visible feature response

$$Y_{ctx} = \mathcal{F}_{FFN}\left(X \otimes M + X\right) \tag{4}$$

where $\otimes$ denotes element-wise multiplication and $\mathcal{F}_{FFN}(\cdot)$ is a feed-forward projection used for channel interaction. This residual calibration preserves the original visible information while emphasizing regions that are structurally consistent with smoke-related morphology.

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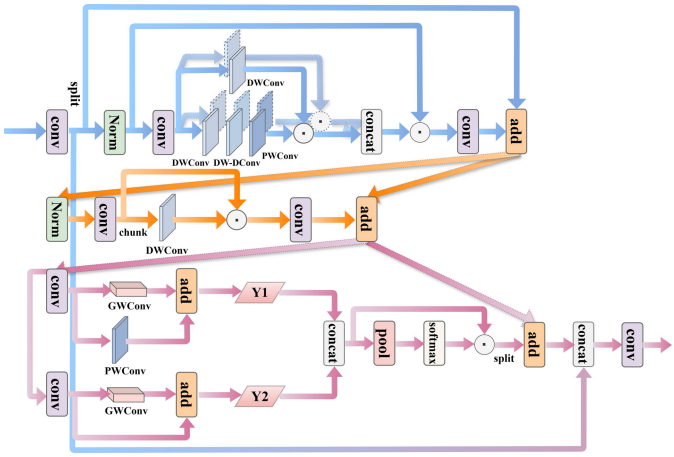


Fig. 4. Network architecture of TAS.

Finally, the local geometry-enhanced feature and the macro-anchor feature are integrated as

$$Y_{gaa} = Y_{geo} + Y_{ctx}. \tag{5}$$

In this design, the geometry-adaptive branch serves as the main path for contour-sensitive smoke representation, while the macro-anchor branch acts as an auxiliary contextual constraint to suppress visually similar background interference. Their coupling enables GAA to produce a smoke-oriented visible representation with improved boundary continuity and structural reliability, which provides a more stable visible prior for subsequent cross-modal interaction.

B. TAS Module

TIR images provide important heat-sensitive cues for fire localization, especially when visible targets are partially obscured by smoke. However, infrared features usually suffer from weak texture expression, blurred object boundaries, and radiation-induced background responses. In forest fire scenes, active fire regions often appear as compact high-response targets, whereas surrounding warm backgrounds and diffused heat responses may introduce confusing activations. To enhance the discriminability of infrared representations, we design the TAS module for the infrared stream, as illustrated in Fig. 4. TAS strengthens salient thermal responses through multiscale contextual encoding, spatial gated refinement, and complementary channel response recalibration.

Given an infrared feature map X_{ir}, TAS first performs multiscale contextual encoding to enlarge the effective perception range of thermal responses. After normalization and point-wise projection, the feature is divided into a context-guidance flow and a response-preserving flow

$$[A, V] = \mathrm{Split}\left(\mathcal{W}_{p1} * \mathrm{LN}\left(X_{ir}\right)\right) \tag{6}$$

where $\mathrm{LN}(\cdot)$ denotes the layer normalization, $\mathcal{W}_{p1}$ is a point-wise projection, and A and V represent the context-guidance flow and the response-preserving flow, respectively.

To describe thermal responses with different spatial extents, the context-guidance flow A is further separated into two groups

$$[A_1, A_2] = \mathrm{Split}\left(A\right). \tag{7}$$

Each group is processed by local and enlarged depth-wise operators. The local operator preserves compact thermal structures, while the enlarged operator captures broader contextual responses

$$T_i = \mathcal{H}_i^e\left(A_i\right) \odot \mathcal{H}_i^l\left(A_i\right), \quad i \in \{1, 2\} \tag{8}$$

where $\mathcal{H}_i^l(\cdot)$ denotes the local depth-wise encoding operator, $\mathcal{H}_i^e(\cdot)$ denotes the enlarged depth-wise contextual operator, and $\odot$ represents element-wise multiplication. The two encoded responses are concatenated to generate a multiscale thermal guidance representation

$$A_{th} = \mathrm{Concat}\left(T_1, T_2\right). \tag{9}$$

The generated thermal guidance representation is then used to recalibrate the response-preserving flow. A point-wise projection maps the gated response back to the original feature space, and a residual formulation is adopted to maintain stable infrared information

$$Y_{mc} = X_{ir} + \boldsymbol{\alpha} \odot \mathcal{W}_{p2} * \left(V \odot A_{th}\right) \tag{10}$$

where $\mathcal{W}_{p2}$ denotes the output projection and $\boldsymbol{\alpha} \in \mathbb{R}^{C\times 1\times 1}$ is a learnable channel-wise residual scaling factor. This stage enhances thermal responses that are consistent across multiple contextual ranges while suppressing isolated background activations.

Although multiscale contextual encoding improves the perception of diffused thermal responses, small fire sources may still suffer from weak boundary contrast. Therefore, TAS further introduces a spatial gated refinement stage to emphasize local salient thermal structures. Specifically, Y_{mc} is projected and divided into a gating flow G and a local response flow R

$$[G, R] = \mathrm{Split}\left(\mathcal{W}_{s1} * \mathrm{LN}\left(Y_{mc}\right)\right). \tag{11}$$

The gating flow is encoded by a depth-wise spatial operator and then used to modulate the local response flow

$$Y_{sg} = Y_{mc} + \boldsymbol{\beta} \odot \mathcal{W}_{s2} * \left(R \odot \mathcal{H}_s\left(G\right)\right) \tag{12}$$

where $\mathcal{H}_s(\cdot)$ denotes the depth-wise spatial encoding operator, $\mathcal{W}_{s2}$ is a point-wise projection, and $\boldsymbol{\beta} \in \mathbb{R}^{C\times 1\times 1}$ is a learnable channel-wise residual scaling factor. This refinement enhances compact thermal anomalies and alleviates boundary weakening caused by excessive contextual smoothing.

To further suppress redundant warm-background responses, TAS performs complementary channel response recalibration. The spatially refined feature Y_{sg} is reduced and split into two channel flows, denoted as F'_{up} and F'_{low}. The first flow is transformed by group-wise and point-wise projections to generate an enhanced response candidate

$$Y_1 = \mathcal{C}_{group}\left(F'_{up}\right) + \mathcal{C}_{point}\left(F'_{up}\right). \tag{13}$$

The second flow preserves more identity information while introducing lightweight group-wise transformation

$$Y_2 = \mathcal{C}_{group}\left(F'_{low}\right) + F'_{low}. \tag{14}$$

Here, Y_1 emphasizes transformed thermal response patterns, while Y_2 retains complementary infrared information with lower transformation disturbance. The two response candidates

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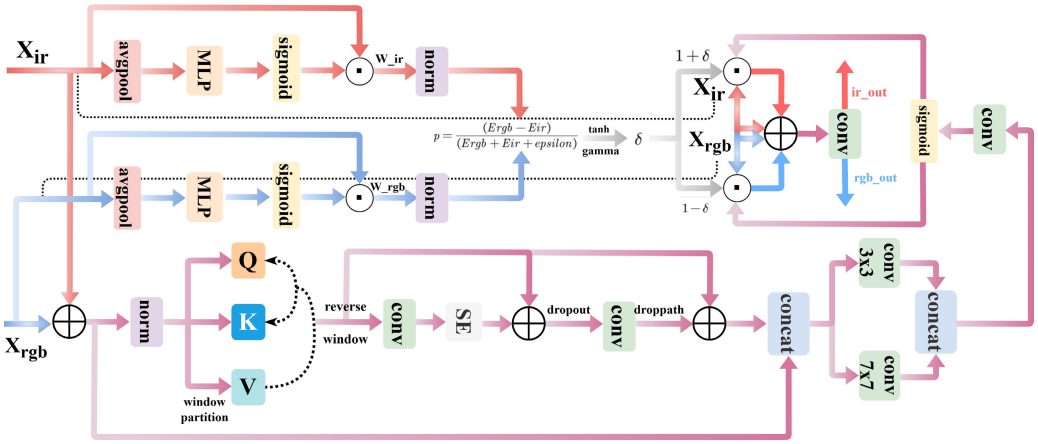


Fig. 5. Detailed architecture of the EGM mechanism.

are concatenated and globally summarized to generate adaptive channel weights

$$S_{fuse} = \text{Concat}\,(Y_1, Y_2) \tag{15}$$

$$\omega = \text{Softmax}\left(\mathcal{P}_{avg}\left(S_{fuse}\right)\right). \tag{16}$$

The recalibrated thermal response is obtained by weighted selection over the two candidate flows

$$Y_{ch} = \sum_{j=1}^{2} \text{Split}_j\left(\omega \odot S_{fuse}\right). \tag{17}$$

Finally, the channel-recalibrated response is integrated with the preceding thermal representation and projected to obtain the TAS output

$$Y_{tas} = \mathcal{W}_o * \text{Concat}\left(Y_{sg}, Y_{ch}\right). \tag{18}$$

Through the above design, the multiscale contextual encoding enlarges the perception range of thermal responses, the spatial gated refinement strengthens compact fire-source structures, and the complementary channel response recalibration adaptively emphasizes informative thermal components. As a result, TAS provides a saliency-enhanced infrared representation that is more robust to texture deficiency, blurred thermal boundaries, and radiation-induced background interference.

C. EGM Mechanism

The smoke-oriented visible representation produced by GAA and the thermal-saliency representation produced by TAS provide complementary cues for forest fire detection. However, their reliability varies significantly with environmental conditions. For example, dense smoke and low illumination may weaken visible responses, whereas solar-heated backgrounds or thermal reflections may introduce unreliable infrared activations. Therefore, directly aggregating the two modality features with fixed weights may propagate degraded responses into the fused representation. To address this issue, we propose the EGM mechanism, as illustrated in Fig. 5. EGM establishes a reliability-aware cross-modal routing scheme, where global modality energy is used to guide local spatial response modulation.

Given the visible and infrared features X_{rgb} and X_{ir}, EGM first computes a scene-level response statistic for each modality to guide adaptive fusion. Specifically, global average pooling and a lightweight channel projection are adopted to generate modality-specific channel confidence G_m

$$G_m = \sigma\left(\text{MLP}\left(\mathcal{P}_{avg}\left(X_m\right)\right)\right), \quad m \in \{rgb, ir\}. \tag{19}$$

The modality energy E_m is defined as the aggregate magnitude of the confidence-weighted feature response

$$E_m = \left\|X_m \odot G_m\right\|_1, \quad m \in \{rgb, ir\}. \tag{20}$$

Here, energy denotes a feature-response statistic computed during forward propagation. The L_1 norm sums the absolute values of the confidence-weighted activations across the channel and spatial dimensions, providing a scalar summary without cancellation between positive and negative responses. This statistic guides cross-modal feature modulation and does not constitute an additional regularization term in the training objective.

Based on the relative energy difference between the two modalities, a global reliability bias is computed as

$$p = \frac{E_{rgb} - E_{ir}}{E_{rgb} + E_{ir} + \epsilon}, \quad \delta = \gamma \tanh\left(p\right) \tag{21}$$

where ϵ is a small constant for numerical stability and γ controls the modulation amplitude. A positive δ indicates a larger confidence-weighted response magnitude in the visible modality and increases its spatial modulation weights, whereas a negative δ increases those of the infrared modality.

To capture local cross-modal spatial dependencies, EGM first constructs an initial fused feature

$$X_{init} = X_{rgb} + X_{ir}. \tag{22}$$

The initial feature is then processed by a window-based contextual interaction block to obtain the contextual response

$$\Phi_{ctx} = \mathcal{F}_{ctx}\left(X_{init}\right) \tag{23}$$

where $\mathcal{F}_{ctx}(\cdot)$ denotes the window-based contextual interaction implemented by the iRMB block. This operation provides local contextual awareness while maintaining computational efficiency.

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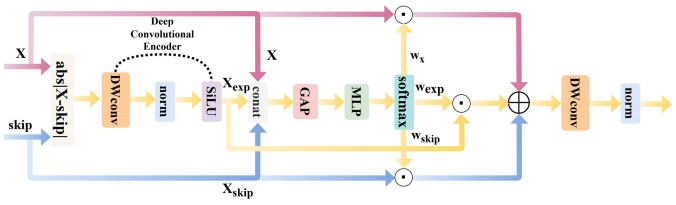


Fig. 6. Overall architecture of the DDR unit.

Subsequently, the initial feature and contextual response are concatenated to generate a joint representation

$$Z = \text{Concat}\left(X_{init}, \Phi_{ctx}\right). \tag{24}$$

A lightweight multiscale pixel modulation unit is then applied to predict modality-specific spatial weights. Specifically, Z is first compressed by a point-wise projection and then processed by two parallel spatial branches with different receptive fields

$$A = \sigma\left(\mathcal{W}_{fuse} * \text{Concat}\left(\mathcal{W}_{3\times3} * \mathcal{W}_r(Z), \mathcal{W}_{7\times7} * \mathcal{W}_r(Z)\right)\right) \tag{25}$$

where $\mathcal{W}_r$ denotes channel reduction, $\mathcal{W}_{3\times3}$ and $\mathcal{W}_{7\times7}$ extract local and enlarged spatial responses, and $\mathcal{W}_{fuse}$ generates a two-channel spatial weight map. The weight map A is split into modality-specific components

$$\left[A_{rgb}, A_{ir}\right] = \text{Split}(A). \tag{26}$$

Finally, the global reliability bias δ modulates the local spatial weights to obtain reliability-aware modality responses

$$\hat{A}_{rgb} = A_{rgb}(1+\delta), \quad \hat{A}_{ir} = A_{ir}(1-\delta). \tag{27}$$

The fused feature is then obtained through residual aggregation

$$Y_{egm} = X_{init} + X_{rgb} \odot \hat{A}_{rgb} + X_{ir} \odot \hat{A}_{ir}. \tag{28}$$

The fused representation is further projected and separated into two modality-specific output flows

$$\tilde{X}_{rgb} = \mathcal{W}_{rgb} * \mathcal{W}_o\left(Y_{egm}\right), \quad \tilde{X}_{ir} = \mathcal{W}_{ir} * \mathcal{W}_o\left(Y_{egm}\right) \tag{29}$$

where $\mathcal{W}_o$ denotes the shared output projection and $\mathcal{W}_{rgb}$ and $\mathcal{W}_{ir}$ are modality-specific heads.

Through this design, EGM couples global confidence-weighted response assessment with local pixel-level modulation. The relative modality response magnitudes provide a scene-dependent cue for adjusting modality contributions, while the local spatial weights enable fine-grained modulation at different locations. Together, these components are designed to reduce interference from degraded modality features and produce adaptive cross-modal representations for subsequent detection.

D. DDR Unit

Although GAA, TAS, and EGM provide robust modality-specific and cross-modal representations, the subsequent multiscale aggregation still faces a critical representation gap. Deep features contain stronger semantic discrimination, but repeated downsampling and contextual abstraction tend to weaken high-frequency details. In contrast, shallow skip features retain richer spatial structures but may contain background noise and weak semantic consistency. Directly fusing these two types of features may cause semantic-detail response conflicts, where deep low-frequency semantics suppress minute fire spots and smoke boundaries preserved in shallow layers. To alleviate this issue, we propose the DDR module, as illustrated in Fig. 6. DDR explicitly extracts cross-layer residual discrepancies and dynamically reallocates the contributions of semantic, detail, and difference-enhanced flows for detail-preserving aggregation.

Given a deep semantic feature X and its corresponding skip feature X_{skip}, DDR first computes their absolute residual to identify the cross-layer response discrepancy

$$X_{diff} = |X - X_{skip}|. \tag{30}$$

The difference map reflects inconsistent responses between the deep semantic flow and the shallow detail flow. These inconsistent responses usually correspond to spatial details weakened during downsampling, such as small fire regions, smoke edges, and local boundary transitions. To enhance such discrepancy cues, a lightweight depth-wise expansion operator is applied to X_{diff}

$$X_{exp} = \sigma\left(\mathcal{B}\left(\mathcal{W}_{dw} * X_{diff}\right)\right) \tag{31}$$

where $\mathcal{W}_{dw}$ denotes the depth-wise convolution, $\mathcal{B}(\cdot)$ denotes the batch normalization, and $\sigma(\cdot)$ represents the SiLU activation function. The obtained X_{exp} serves as a difference-enhanced flow that complements the semantic and skip features with localized structural discrepancy information.

To avoid rigidly combining features from different semantic levels, DDR introduces a dynamic multiplexing strategy over three complementary flows: the deep semantic flow X, the shallow detail flow X_{skip}, and the difference-enhanced flow X_{exp}. These three flows are concatenated along the channel dimension

$$Z = \text{Concat}\left(X, X_{skip}, X_{exp}\right). \tag{32}$$

Then, global average pooling is used to summarize the overall response distribution

$$s = \mathcal{P}_{avg}(Z). \tag{33}$$

A lightweight two-layer projection is further adopted to estimate the relative importance of the three flows

$$E = \mathcal{W}_{out} * \text{ReLU}\left(\mathcal{W}_{in} * s\right). \tag{34}$$

The dynamic allocation weights are generated by a Softmax function

$$\left[w_x, w_{skip}, w_{exp}\right] = \text{Softmax}(E) \tag{35}$$

where w_x, w_{skip}, and w_{exp} denote the adaptive weights assigned to the semantic, detail, and difference-enhanced flows, respectively. This competitive allocation allows DDR to adaptively adjust the fusion tendency according to the current feature distribution, rather than using a fixed aggregation rule.

Finally, the three flows are softly combined and projected to produce the reconstructed feature

$$Y_{ddr} = \mathcal{B}\left(\mathcal{W}_{redu} * \left(w_x \odot X + w_{skip} \odot X_{skip} + w_{exp} \odot X_{exp}\right)\right) \tag{36}$$

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where $\mathcal{W}_{redu}$ denotes a 1×1 projection used for feature reduction and interaction. Through this design, DDR converts rigid cross-layer fusion into discrepancy-aware dynamic aggregation. The semantic flow maintains high-level discriminability, the skip flow preserves spatial details, and the difference-enhanced flow compensates for local responses weakened by semantic abstraction. Consequently, DDR improves the preservation of minute fire regions and smoke boundaries during multiscale aggregation.

IV. Experiments and Results

A. Datasets

To objectively evaluate model performance, rigorous spatial registration and precise custom annotation are performed based on two authoritative public aerial forest fire datasets, namely FLAME 2 and FLAME 3 to establish our experimental benchmark.

1) *FLAME 2 Dataset [42]:* Acquired during a 2021 prescribed burn in an Arizona pine forest, FLAME 2 was recorded by a dual-camera UAV at 30 fps and contains 53 541 paired visible and infrared frames extracted from seven raw videos. The visible images have resolutions of 3840×2160 or 1920×1080, whereas the infrared frames are 640×512. Although FLAME 2 provides cropped and scaled RGB–thermal pairs, their resolution is limited to 254×254, which is insufficient for fine-grained fire detection. We therefore reregister the original high-resolution images by matching cross-modal feature points within the overlapping field of view and estimating the corresponding spatial transformation matrix. The visible images are then warped according to the estimated transformation and cropped to remove nonoverlapping regions, after which both modalities are normalized to 640×512. Raw samples also include expert binary annotations, where smoke labels require a 50% visual area threshold, together with weather data and a georeferenced point cloud with a spatial resolution of 7.6 cm per pixel.
2) *FLAME 3 Dataset [43]:* Collected during a 2023 prescribed burn in Oregon, FLAME 3 contains 738 image quartets, including 622 fire and 116 background samples. Each quartet comprises a 4000×3000 raw RGB image, a field-of-view-corrected RGB image, a raw thermal image, and a radiometric thermal TIFF image, with the latter three provided at a resolution of 640×512. In this study, we use the officially provided corrected RGB images rather than the raw RGB images.

Considering the severe visual repetition among adjacent frames in continuous drone recordings, a frame interval sampling strategy is adopted to reduce information redundancy and enrich training scenarios. Through this approach, 1280 distinct image pairs are ultimately extracted from FLAME 2 and 560 pairs are selected from FLAME 3. Finally, the processed datasets are randomly divided into training, validation, and testing sets using a 7 to 2 to 1 ratio.

TABLE I
Training Hyperparameter Sensitivity of AAFNet

ID	lr_0	lr_f	Momentum	mAP @50	mAP @50–95
1	**0.01**	**0.01**	**0.937**	**75.2**	**52.1**
2	0.005	0.01	0.937	74.3	51.5
3	0.05	0.01	0.937	75.1	52.0
4	0.03	0.01	0.937	74.7	51.9
5	0.003	0.01	0.937	73.0	51.1
6	0.01	0.02	0.900	74.1	51.7
7	0.01	0.02	0.970	74.8	51.9
8	0.01	0.01	0.970	75.0	51.9
9	0.01	0.005	0.970	75.2	52.0
10	0.01	0.005	0.900	74.6	51.7

* Note: Detection metrics are reported in percentages (%). The reference configuration is shown in bold.

B. Experimental Setup

All experiments are conducted on an Ubuntu 22.04 workstation with an Intel Core i7 14700KF processor and an NVIDIA GeForce RTX 4080 SUPER GPU. We use SGD with an initial learning rate of 0.01, a final learning-rate ratio of 0.01, and a momentum of 0.937, giving a scheduled final learning rate of 10^{-4}. These values match those reported for YOLOv9 and YOLOv10 [44], [45]. Models are trained for 300 epochs with a batch size of 16 and an input resolution of 640×512 pixels. These settings are shared across comparative experiments, except for the hyperparameter variations examined next.

Training Hyperparameter Sensitivity: Table I compares ten configurations with all other settings fixed. Configurations 1–5 vary lr_0 at $lr_f = 0.01$ and momentum = 0.937, while configurations 1 and 6–10 evaluate the combinations of lr_f and momentum at $lr_0 = 0.01$. The reference configuration achieves the highest mAP@50–95 of 52.1%, and increasing lr_0 to 0.03 or 0.05 does not incur further improvement.

A larger lr_f retains a higher late-stage learning rate, while larger momentum increases the persistence of past gradients. Since the learning rate scales the momentum-based update, their combined effects influence convergence behavior. At $lr_0 = 0.01$, mAP@50–95 varies from 51.7% to 52.1% across the evaluated combinations. The epoch-wise training losses show overall convergence across all ten configurations, without sustained divergence or persistent large-amplitude oscillations. These observations support the setting on the current dataset within the tested parameter range.

C. Ablation Experiments

To validate the effectiveness of the proposed core modules (GAA, TAS, EGM, and DDR), a progressive ablation study is conducted under identical training settings, as detailed in Table II. In addition to detection accuracy and parameter count, computational cost, inference time, and peak GPU memory usage are reported. All inference measurements are performed on the same NVIDIA GeForce RTX 4080 SUPER GPU using a batch size of 1, FP32 precision, and an input resolution of 640×512. GFLOPs measures the computational cost per

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TABLE II
ABLATION STUDY RESULTS FOR KEY MODULES OF AAFNET

Methods	GAA	TAS	EGM	DDR	mAP @50	mAP @50-95	AP@50 (fire)	AP@50 -95(fire)	AP@50 (smoke)	AP@50 -95(smoke)	Precision	Recall	Params	GFLOPs	Speed	Peak Memory
Baseline					72.4	50.2	68.0	37.5	76.8	63.0	77.9	67.4	2.463	5.42	2.45	166.66
+GAA	✓				73.4	50.8	67.8	37.3	78.9	64.3	78.0	67.4	2.391	5.31	3.00	166.56
+TAS	✓	✓			74.0	50.9	68.7	38.0	79.3	63.8	78.7	66.82	2.345	5.21	3.54	166.46
+EGM	✓	✓	✓		74.4	51.7	68.8	37.9	80.0	65.5	80.4	65.7	2.2705	5.39	4.41	197.47
AAFNet	✓	✓	✓	✓	**75.2**	**52.1**	**70.5**	**39.5**	**80.0**	**64.8**	**80.1**	**68.4**	**2.531**	**5.87**	**5.04**	**202.62**

* Note: Detection metrics are reported in percentage (%), and Params is reported in millions (M). Speed and Peak Memory are reported in ms and MB, respectively.

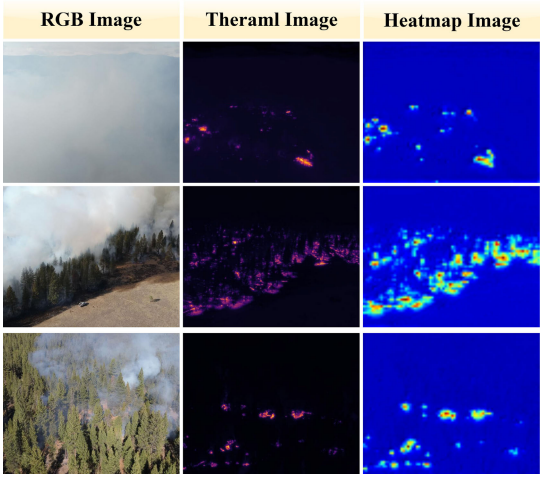


Fig. 7. Visualization demonstrating the effectiveness of the proposed TAS module.

paired RGB–IR input. Speed denotes the average forward-inference time per image pair in milliseconds, excluding preprocessing and postprocessing, and Peak Memory denotes the peak GPU memory usage during inference in MB.

1) *Effectiveness of GAA and TAS:* As shown in Table II, introducing the GAA module enables the baseline model to fit the nonrigid boundaries of smoke, increasing the smoke AP@50 from 76.8% to 78.9%. Furthermore, adding the TAS module improves the fire AP@50% to 68.7% and raises the overall mAP@50% to 74.0%. This significant enhancement in fire detection is directly attributed to the hierarchical wide-range perception operators within the TAS module. By capturing long-range thermal gradients and explicitly eliminating background radiation artifacts, this mechanism successfully penetrates heavy smoke interference to pinpoint the underlying fire cores. As corroborated by the feature heatmaps in Fig. 7, TAS achieves strict spatial topological consistency with actual heat sources. Both quantitative metrics and qualitative visualizations demonstrate the synergistic advantage of this architectural division, where GAA specializes in extracting smoke features and TAS focuses on localizing the fire core. Meanwhile, the configuration incorporating both GAA and TAS reduces computational cost from 5.42 to 5.21 GFLOPs, maintains nearly unchanged peak GPU memory usage at 166.46 MB, and achieves a forward-inference time of 3.54 ms on the tested GPU.

TABLE III
COMPARISON OF DIFFERENT FUSION MODULES WITHIN AAFNET

Fusion module	mAP@50(%)	mAP@50–95(%)	Precision(%)	Recall(%)
CIFusion [46]	74.4	51.3	79.3	67.2
MFMFusion [47]	74.2	51.5	79.4	66.9
DECA+DEPA [34]	73.9	51.3	78.0	67.5
CFTFusion [48]	74.3	51.0	79.2	67.3
MASAGFusion [49]	74.1	51.2	79.1	67.6
EGM (Ours)	**75.2**	**52.1**	**80.1**	**68.4**

2) *Effectiveness of EGM:* Adding the EGM increases precision to 80.4%. To address dynamic signal-to-noise ratio inversions, EGM dynamically allocates modality weights by quantifying instantaneous energy responses. This active regulation suppresses background interference and ensures feature purity during fusion. Consequently, the more stringent smoke AP@50–95 rises to 65.5%, while parameters decrease to 2.27 M. The configuration incorporating GAA, TAS, and EGM requires 5.39 GFLOPs, which is lower than the baseline's 5.42 GFLOPs and achieves a forward-inference time of 4.41 ms with a peak GPU memory usage of 197.47 MB on the tested GPU.

To further evaluate the fusion design, EGM in the complete AAFNet is replaced with five alternative fusion modules while keeping the remaining architecture and training settings unchanged. As shown in Table III, EGM performs best across all four metrics, improving mAP@50 and mAP@50–95 by 0.8 and 0.6 percentage points over the strongest alternative for each metric, respectively. EGM derives a global modality bias from confidence-weighted feature-response magnitudes and adopts it to adjust local spatial weights, combining global modality weighting with local feature selection. These results support the effectiveness of this design in integrating complementary visible and infrared information.

To assess EGM under temporal distribution changes, we perform a chronological split of the annotated RGB–IR pairs from FLAME 2 and the FLAME 3 CV subset. Samples are ordered using source-video frame indices for FLAME 2 and acquisition timestamps for FLAME 3, and the first 70%, subsequent 20%, and final

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TABLE IV
ABLATION STUDY RESULTS FOR EGM UNDER RANDOM AND CHRONOLOGICAL SPLITS

Split	Method	Overall Performance		Class-wise AP@50(%)	
		mAP@50(%)	mAP@50–95(%)	Fire	Smoke
Random	Baseline+GAA+TAS	74.0	50.9	68.7	79.3
Random	Baseline+GAA+TAS+EGM	**74.4**	**51.7**	**68.8**	**80.0**
Chronological	Baseline+GAA+TAS	70.8	49.6	65.2	76.4
Chronological	Baseline+GAA+TAS+EGM	**71.3**	**50.3**	**65.4**	**77.2**

10% are adopted for training, validation, and testing, respectively. We retrain Baseline+GAA+TAS with and without EGM, both excluding DDR, under identical settings. As shown in Table IV, EGM improves mAP@50 from 70.8% to 71.3% and mAP@50–95 from 49.6% to 50.3%, corresponding to gains of 0.5 and 0.7 percentage points. These gains are comparable to the corresponding improvements of 0.4 and 0.8 percentage points under the random split. Smoke AP@50 upgrades from 76.4% to 77.2%, while fire AP@50 increases from 65.2% to 65.4%. These results indicate that EGM retains its detection benefits under temporal changes, with a larger contribution to smoke detection.

3) *Effectiveness of DDR:* Integrating the DDR yields an optimal overall mAP@50 of 75.2%. To resolve the loss of high-frequency boundaries caused by deep semantic abstraction, DDR explicitly reconstructs residual differences between adjacent feature hierarchies to compensate for fine-grained edge details. This surges the fire AP@50% to 70.5% and significantly restores recall to 68.4%, breaking the inherent tradeoff between precision and recall.

To quantify the additional overhead introduced by DDR, we compare the Baseline+GAA+TAS+EGM configuration with the complete AAFNet in Table II. Adding DDR introduces 0.2605 M parameters and 0.48 GFLOPs, increasing the computational cost from 5.39 to 5.87 GFLOPs (approximately 8.9%). The measured peak GPU memory usage increases by 5.15 MB, from 197.47 to 202.62 MB (approximately 2.6%), while forward-inference latency rises from 4.41 to 5.04 ms on the RTX 4080 SUPER. These additional costs accompany improvements of 0.8 and 0.4 percentage points in mAP@50 and mAP@50–95, respectively, indicating a favorable accuracy–resource tradeoff under the evaluated conditions.

Cai et al. [50] report a 24.3-ms inference latency for EfficientViT-B1 (4.8 M parameters) on the Jetson AGX Orin using TensorRT FP16, providing a practical example of real-time edge deployment for lightweight vision models. AAFNet has a smaller parameter count of 2.531 M and requires 5.87 GFLOPs per RGB–IR image pair, with a measured peak GPU memory usage of 202.62 MB on our evaluation platform. Together, this published deployment example and our measured resource profile support the potential of AAFNet for real-time UAV edge inference with hardware-adapted execution. In particular, DDR improves detection performance with merely 5.15 MB of additional peak GPU memory, preserving the model's compact resource footprint and its potential for deployment on resource-constrained platforms.

D. Comparison With SOTA Methods

To comprehensively validate the superiority of our model, it is evaluated against mainstream single-modal detectors and recent multimodal fusion algorithms under identical experimental settings (Table V). Since single-modal YOLO series detectors and the forest fire-specific PDENet are constrained by relying solely on RGB or IR modalities, they struggle to balance smoke penetration and morphological extraction, achieving a maximum mAP@50 of merely 61.3% (RGB) and 69.7% (IR). They quickly hit a performance ceiling under complex wildland background interference.

While existing multimodal fusion methods improve upon single-modal baselines, they still exhibit critical bottlenecks. CDC-YOLOFusion and DEYOLO employ cross-modal enhancement, yet their inability to decouple heterogeneous physical properties under dynamic signal-to-noise ratio inversions causes their mAP@50 to stagnate at 72.1% and 72.0%, respectively. Furthermore, methods relying on global interactions or deep semantic abstraction fail to suppress forest fire-specific thermal radiation noise and tend to blur fragile microfire boundaries. Consequently, ICAFusion suffers from massive background false alarms, dropping its precision to 75.8%. Similarly, Fusion-DETR and EI2Det struggle with fine-grained texture loss, yielding suboptimal mAP@50–95 scores of 48.8% and 49.6%.

To thoroughly break these bottlenecks, our architecture is explicitly tailored to the physical properties of forest fires. The GAA and TAS modules decouple modalities at the source, where GAA adaptively fits nonrigid smoke boundaries to restore spatial sensitivity, while TAS eliminates thermal radiation artifacts via wide-range perception. To handle complex environmental interference, the EGM mechanism dynamically allocates modality weights based on instantaneous energy responses, effectively suppressing background noise and boosting overall precision. Finally, the DDR unit explicitly reconstructs residual differences between feature hierarchies to preserve the fine-grained textures of early stage fires, overcoming the recall bottleneck caused by deep feature loss.

Ultimately, our model achieves state-of-the-art comprehensive performance, yielding 75.2% mAP@50 and 52.1%

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TABLE V
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE MERGED FLAME 2 AND FLAME 3 DATASETS

Methods	Pub-year	Modality	mAP@50 (%)	mAP@50-95 (%)	Precision (%)	Recall (%)	Params (M)
YOLOv5n	2020	RGB	57.7	40.8	62.7	55.1	2.503
YOLOv8n	2023	RGB	59.4	42.1	66.5	53.8	3.006
YOLOv11 [51]	2024	RGB	58.2	41.2	65.9	52.4	2.582
PDENet [16]	2025	RGB	61.3	42.5	67.2	56.6	4.407
YOLOv5n	2020	IR	69.5	45.3	74.5	62.5	2.503
YOLOv8n	2023	IR	69.6	45.6	74.7	63.6	3.006
YOLOv11 [51]	2024	IR	69.5	45.2	74.4	63.8	2.582
PDENet [16]	2025	IR	69.7	45.8	74.9	64.1	4.407
CDC-YOLOFusion [52]	2024	RGB+IR	72.1	50.5	77.8	67.8	153.6
DEYOLO [34]	2024	RGB+IR	72.0	50.2	77.4	67.1	74.7
EI2Det [53]	2025	RGB+IR	71.1	49.6	76.6	66.2	122.6
ICAFusion [31]	2024	RGB+IR	70.8	49.0	75.8	65.4	114.7
Fusion-DETR [54]	2025	RGB+IR	70.1	48.8	75.2	65.0	34.4
Ours	-	**RGB+IR**	**75.2**	**52.1**	**80.1**	**68.4**	**2.531**

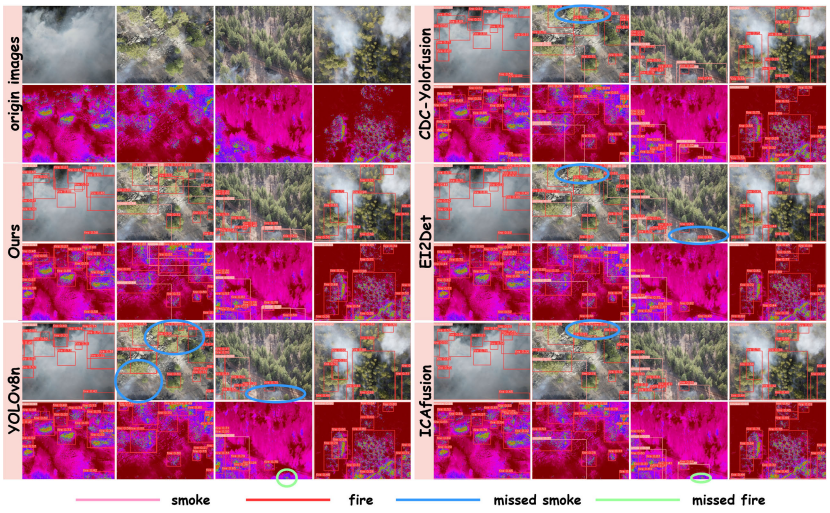


Fig. 8. Qualitative comparison of detection results among different methods on the FLAME 2 dataset.

mAP@50–95 and 80.1% precision, and 68.4% recall. Notably, it accomplishes this with only 2.531 M parameters—a mere fraction of the 153.6 M required by CDC-YOLOFusion or the 114.7 M by ICAFusion. This extreme parameter efficiency, coupled with superior detection accuracy, supports its potential for real-time edge deployment on edge computing platforms.

E. Visualization Results

As shown in Figs. 8 and 9, the qualitative results on the FLAME 2 and FLAME 3 datasets highlight the differences among the compared methods. Pink and red bounding boxes indicate smoke and fire regions. Blue and green circles highlight missed smoke and fire targets. The wildland testing scenarios contain dense canopy occlusions and extreme scale variations. In these complex environments, baseline models exhibit severe limitations due to their specific architectural designs. YOLOv8n employs a naive dual-modal fusion strategy. However, it lacks cross-modal alignment and modality-specific feature decoupling. Meanwhile, direct fusion of heterogeneous features triggers thermal noise interference. Consequently, YOLOv8n fails to detect canopy-obscured smoke and misses distant microfires. CDC-YOLOFusion utilizes cross-scale dynamic convolutions to extract bimodal features. This mechanism lacks explicit priors for nonrigid morphology and misses canopy-obscured smoke regions in the second and third examples of Fig. 8. EI2Det employs an edge-guided fusion module designed for autonomous driving. This rigid edge guidance smooths out the fragile boundaries of forest microtargets. Consequently, EI2Det misses discrete microfires and amorphous smoke across both datasets. ICAFusion introduces cross-attention transformers for global feature interaction. This global modeling struggles to fit the local amorphous diffusion of low-contrast targets. As a result, ICAFusion overlooks the large-scale smoke plume in the second example of Fig. 9. These failures stem from ignoring the unique physical properties of wildland fires. In contrast, our proposed AAFNet alleviates these defects and achieves more complete target detection. The GAA module adaptively fits the nonrigid boundaries of smoke via decoupled deformable sampling. The TAS module captures long-range thermal gradients to localize distinct thermal signatures. The DDR unit explicitly reconstructs residual differences to compensate for lost

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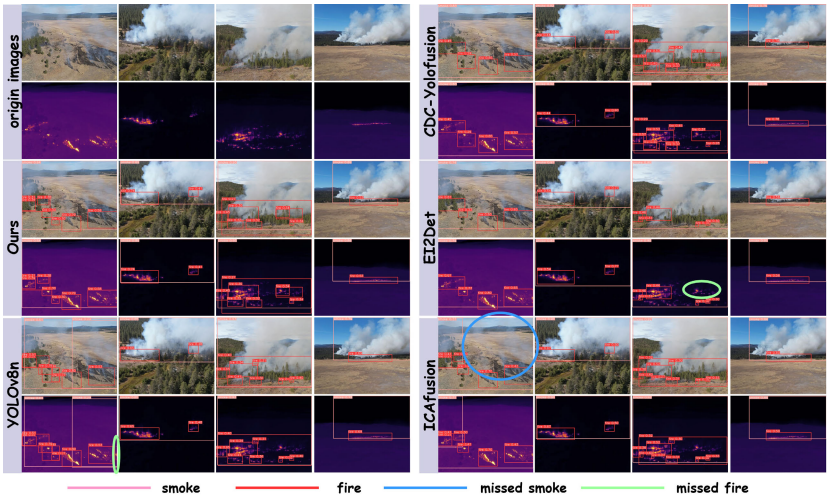


Fig. 9. Qualitative comparison of detection results among different methods on the FLAME 3 dataset.

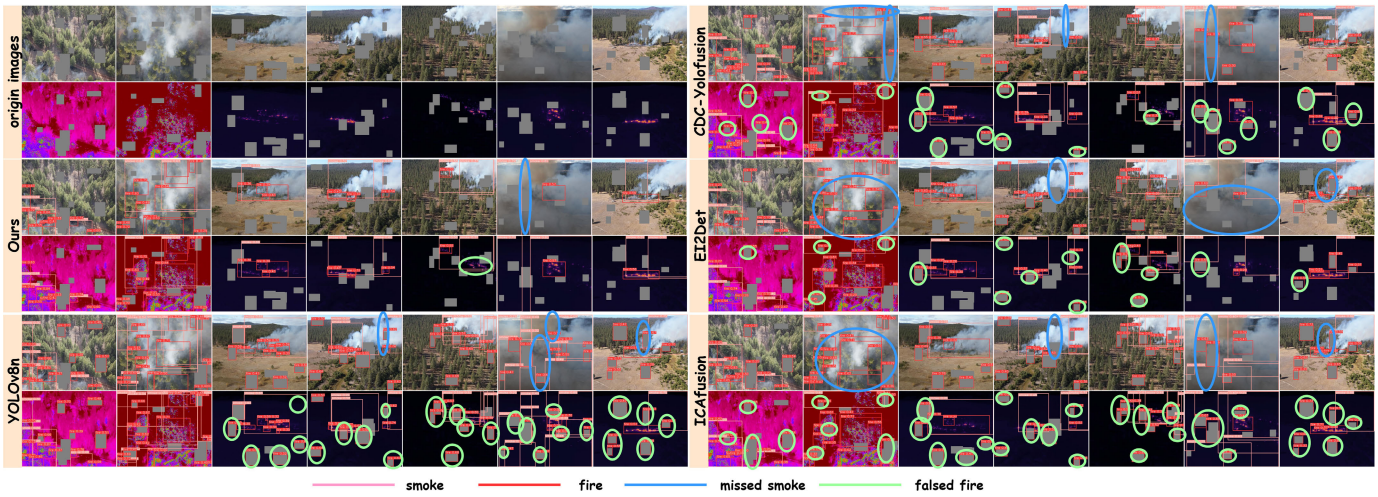


Fig. 10. Qualitative comparison of different detection methods under simulated extreme occlusion.

high-frequency edge details of microfires. The EGM actively suppresses background interference to ensure feature purity. Through these targeted mechanisms, our AAFNet prevents omissions and yields robust detections in diverse wildland scenarios.

F. Robustness Evaluation Under Occlusion

We construct a synthetic occlusion dataset with random rectangular masks to evaluate model robustness under severe localized information loss. For each spatially aligned RGB–IR pair, identical masks are applied at the same locations in both modalities, with their positions and sizes randomly generated. Masked pixels are replaced with a constant gray value to remove local visual and thermal observations. This operation provides a controlled stress test of observation degradation rather than a reproduction of a specific sensor-failure mechanism. The original ground-truth labels remain unchanged, preserving a common reference for assessing whether originally annotated targets remain detectable after masking. As shown in Fig. 10, pink and red boxes indicate smoke and fire detections, while blue and green circles highlight missed smoke targets and false fire detections.

The qualitative comparisons in Fig. 10 show missed smoke targets and false fire detections among the compared methods under the tested masking conditions. Several false fire detections occur near masked regions, suggesting sensitivity to artifacts introduced by localized information loss. CDC-YOLOFusion, EI2Det, and ICAFusion also exhibit noticeable detection errors when corresponding RGB and infrared regions simultaneously lose useful observations.

In contrast, AAFNet exhibits fewer apparent false fire detections while retaining detections of several partially occluded targets in the illustrated examples. The model operates on the remaining observations without explicitly classifying the source of degradation. Asymmetric feature encoding preserves modality-specific cues, EGM adjusts modality contributions according to confidence-weighted feature-response magnitudes, and DDR enhances structural information retained across feature hierarchies. This design supports the use of remaining target cues and contextual information under partial occlusion. These qualitative results suggest improved

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TABLE VI
PERFORMANCE AND PARAMETER EFFICIENCY ACROSS DIVERSE ENVIRONMENTAL DATASET

Methods	mAP @50	mAP @50-95	Prec.	Rec.	Params
YOLOv8n	95.6	63.9	92.3	91.8	2.463
CP-YOLOv11-MF	96.3	62.9	92.5	93.5	23.0
CDC-YOLOFusion	96.1	64.2	92.6	92.0	153.6
DEYOLO	96.1	64.1	92.4	92.1	74.7
EI2Det	95.9	63.9	92.1	91.9	122.6
ICAFusion	95.8	63.8	91.9	91.7	114.7
Fusion-DETR	95.8	63.9	91.7	91.8	34.4
Ours	**96.3**	**65.2**	**92.9**	**92.2**	**2.531**

* Note: Metrics in %, Params in millions (M).
Prec.: Precision; Rec.: Recall.

robustness to the tested localized information loss, without establishing explicit discrimination between sensor malfunction and physical occlusion.

G. Architectural Generalizability Across Diverse Distributions

To further assess the generalizability of our AAFNet, the evaluation is extended to the recently introduced RGBT-3M dataset [55]. Given the scarcity of large-scale, high-quality dual-modality data in this field, this collection provides a vital platform to test AAFNet against diverse real-world environmental shifts, as shown in Table VI. In this large-scale setting, AAFNet achieves an mAP@50 of 96.3%, matching the accuracy of the original CP-YOLOv11-MF baseline while maintaining a significant lead in efficiency. Specifically, the architecture delivers this performance with approximately 1/60th of the computational parameters required by heavyweight frameworks like CDC-YOLOFusion, further validating the inherent efficiency of our asymmetric encoding and EGM. Moreover, our AAFNet reaches a state-of-the-art mAP@50–95 of 65.2%, outperforming the baseline by 2.3 percentage points. Such a margin in precise regression highlights the sustained effectiveness of the DDR unit and GAA module in preserving fine-grained structural details across varying data distributions. These results collectively reinforce the practical reliability and robustness of AAFNet for deployment on resource-constrained UAV platforms.

V. CONCLUSION

We introduce an AAFNet for multimodal remote sensing forest fire detection. The architecture integrates four pivotal components to decouple heterogeneous physical properties and preserve high-frequency edge details. At the extraction source, the GAA module fits nonrigid smoke boundaries via decoupled deformable sampling. In parallel, the TAS module captures long-range thermal gradients to suppress radiation artifacts. During feature fusion, the EGM mechanism quantifies instantaneous energy responses to dynamically allocate modality weights. This improves feature purity during fusion under dynamic signal-to-noise ratio inversions. Finally, the DDR unit explicitly reconstructs residual differences between adjacent hierarchies. This prevents the smoothing of microtargets by deep semantic abstraction.

Extensive experiments on complex forest fire datasets validate our AAFNet. The framework achieves a 75.2% mAP@50, a 52.1% mAP@50–95, and a 68.4% recall, comprehensively outperforming existing multimodal detectors. Moreover, synthetic occlusion experiments verify the robustness of AAFNet under severe data loss, while additional evaluation on an independent RGB-T dataset further demonstrates its generalization capability across diverse environmental distributions. Crucially, the model requires merely 2.531 M parameters. This compact resource footprint supports the potential of AAFNet for efficient deployment while maintaining strong detection performance.

Nevertheless, the current AAFNet displays certain limitations. Performance degradation may occur under extreme meteorological conditions or within unseen forest topographies. Additionally, achieving sustained real-time inference on severely resource-constrained micro aerial vehicles remains challenging. Future research will focus on improving cross-domain generalization across diverse geographical datasets. For lightweight deployment, we will investigate branch-aware structured channel pruning, where the visible and thermal streams are compressed according to their respective feature redundancy while maintaining compatible channel dimensions at cross-modal interaction stages. Modality-aware knowledge distillation will further be explored to preserve the specialized representations of the two branches and the cross-modal interactions learned by the full model. Hardware-level optimization will also be considered to improve practical deployment efficiency.

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